

2018 Ph H2 Q13

Section: Particles and Waves

Topic: Spectra

An emission line spectrum is produced when a gas at low pressure is heated.

(a) Explain how an emission line spectrum is produced.

✓ When atoms in a low-pressure gas are **excited** (e.g. by heat or electricity), electrons absorb energy and move to **higher energy levels**. When these electrons **fall back** to lower energy levels, they **emit photons** of specific energies (and hence, specific wavelengths), producing **bright lines** on a dark background — the emission spectrum.

(b) Explain how an absorption spectrum is produced.

✓ When **white light** passes through a **cool gas**, electrons in the gas **absorb photons** of specific energies corresponding to the energy gaps between atomic levels. These absorbed wavelengths appear as **dark lines** in the otherwise continuous spectrum.

(c) The emission line spectrum from a distant galaxy is observed.

The spectral lines are **shifted** compared to the same lines observed in a laboratory. State what this observation indicates about the galaxy's motion relative to Earth.

✓ The **shift in spectral lines** (usually a **redshift**) indicates that the galaxy is **moving away** from Earth.

If the lines are shifted toward longer wavelengths (red end of spectrum), it shows the galaxy is **receding**.

Revision Tips

- **Emission spectra** = electrons **falling** → photons emitted
- **Absorption spectra** = electrons **absorbing** photons → dark lines
- Redshift in spectra = object is **moving away**
- Lines shift due to the **Doppler effect** on light